

Graph Algorithms: Convenient Problem Transformations

Design and Analysis of Algorithms

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Formulating Shortest Path Problems: Change the Problem, not the Algorithm...

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- Suppose some roads charge a \$1 toll. How do we find a shortest path if we can pay at most \$3 worth of tolls?
- What if every edge e has an associated failure probability $f(e)$, and we'd like to find an s - t path of maximum reliability?

$$\text{Prob}[e \text{ remains intact}] = r(e) = 1 - f(e)$$

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- What if you have capacities governing the amount of flow that can go through nodes, rather than edges?